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SpectriPy: Enhancing Cross-Language Mass Spectrometry Data Analysis with R and Python

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Figure 1: SpectriPy package logo

Summary

Mass spectrometry (MS) is a key technology used across multiple fields, including biomedical research and life sciences. The data is often times large and complex, and analyses must be tailored to the experimental and instrumental setups. Excellent software libraries for such data analysis are available in both R and Python, including R packages from the RforMassSpectrometry initiative such as *Spectra*, *MsCoreUtils*, *MetaboAnnotation*, and *CompoundDb* (Rainer et al., 2022), as well as Python libraries like *matchms* (Huber et al., 2020), *spectrum_utils* (Bittremieux, 2020), *Pyteomics* (Goloborodko et al., 2013) and *pyOpenMS* (Röst et al., 2014). The *reticulate* R package (Ushey et al., 2025) provides an R interface to Python enabling interoperability between the two programming languages. The open-source *SpectriPy* R package builds upon *reticulate* and provides functionality to efficiently translate between R and Python MS data structures. It can convert between R's `Spectra::Spectra` and Python's `matchms.Spectrum` and `spectrum_utils.spectrum.MsmsSpectrum` objects and

includes functionality to directly apply spectral similarity, filtering, normalization, etc. routines from the Python *matchms* library on MS data in R. *SpectriPy* hence enables and simplifies the integration of R and Python for MS data analysis, empowering data analysts to benefit from the full power of algorithms in both programming languages. Furthermore, software developers can reuse algorithms across languages rather than re-implementing them, enhancing efficiency and collaboration.

Statement of need

Over the past decade, tremendous efforts have been made to develop powerful algorithms and excellent data analysis software for MS data analysis. Each of these software covers different and in part complementary aspects in the analysis of MS data, but their integration into a single workflow remains a challenge, in particular across programming languages. To avoid the need for repeated implementation of algorithms in different programming languages we developed the *SpectriPy* package. By leveraging R's *reticulate* package, and translating between R and Python MS data structures, this package enables seamless cross-language integration of MS data analysis algorithms within unified analysis workflows.

Description

Reproducible examples and use case analyses enabled by *SpectriPy* can be found in the package's vignette and the Metabonaut resource (Louail & Rainer, 2025). In this paper, we focus on the technical details and features of the package.

Installation

During installation, *SpectriPy* automatically configures a Python environment and installs all required libraries. The installation can be configured through several environment variables that also allow users to disable the automatic setup and instead use an available Python environment of the host system. Detailed installation instructions can be found in the package's GitHub repository and in the package's vignette.

Translating MS data objects between R and Python

As its core functionality, *SpectriPy* allows translation between R and Python MS data structures. In particular, *SpectriPy* provides the functions `rspec_to_pyspec()` and `pyspec_to_rspec()` to convert between R's `Spectra::Spectra` and Python's `matchms.Spectrum` and `spectrum_utils.spectrum.MsmsSpectrum` objects. These functions also handle the conversion, and any required renaming and reformatting of spectra metadata, such as MS level, retention times, or any other arbitrary metadata available in the MS data object. For more efficient integration of Python MS data objects into R, *SpectriPy* implements a dedicated backend class for `Spectra::Spectra` objects. Such backend classes handle the MS data for `Spectra::Spectra` objects. Through different backend implementations, `Spectra::Spectra` can gain support for additional data and file formats or memory-efficient on-disk or remote data storage modes. *SpectriPy*'s `MsBackendPy` backend keeps only a reference to the original data object in Python and retrieves and translates MS data, or subsets thereof, only upon request from that object. This enables seamless and memory-efficient integration of Python MS data objects into R for more powerful cross-language analysis workflows. An example of such a combined R-Python data analysis workflow, which can be realized e.g. using the Quarto system, is provided in the following code snippets. In the Python code block below, MS data are imported and processed.

```
#' Python session:
#'  Import data and perform initial processing
```

```python
import matchms
import matchms.filtering as mms_filt
from matchms.importing import load_from_mgf
mgf_py = list(load_from_mgf(<MGF file>))

#'  Scale intensities
for i in range(len(mgf_p)):
  mgf_py[i] = mms_filt.normalize_intensities(mgf_py[i])
```

To continue the analysis in R, a `Spectra::Spectra` object with a `MsBackendPy` backend class is created, referring to the Python data object defined in the associated Python session. All data from this data object is accessible in R, with the entire or subsets of the data translated on-the-fly upon request. This strategy ensures memory efficiency and minimizes the number of data copies.

```r
#' R session:
#'  Create an R data object for the MS data in the associated Python session
library(Spectra)
library(SpectriPy)
sps <- Spectra("mgf_py", source = MsBackendPy())

#'  Retrieve the MS peaks data for the 1st spectrum
peaksData(sps[1])
```

Integrated functionality from the *matchms* Python library

SpectriPy's `compareSpectriPy()` and `filterSpectriPy()` functions allow spectra comparison, and filtering and processing routines, respectively, from the *matchms* Python library to be called directly from R. These functions internally translate the MS data from a `Spectra::Spectra` object to the respective Python MS data structures, execute the Python functions, and collect and convert the results to R data types, enabling the integration of functionality from the *matchms* Python library directly into R-based analysis workflows.

Perspective

SpectriPy started as a collaboration of R and Python developers, with the latest contributions added during the EuBIC-MS Developers Meeting in 2025. Collaborative development will be further encouraged to extend *SpectriPy* with additional functionality, support for additional libraries, and data structures. New use cases will be integrated into larger interactive tutorial frameworks such as the Metabonaut resource (Louail & Rainer, 2025), enabling users to seamlessly integrate R and Python into their MS data analysis pipelines.

Ultimately, the long-term goal is to promote cross-language compatibility and reproducibility in computational mass spectrometry. By leveraging the strengths of both R and Python, *SpectriPy* will help develop flexible and efficient MS data analysis workflows, reduce redundancy and promote innovation in the field.

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